LESSON 8.2

Write and Solve Equations Using Addition or Subtraction

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve one-step addition and subtraction equations using inverse operations.

WHY THIS MATTERS Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case. The equation is n + 23 = 58. What is n?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Equation *Spanish: Ecuación*	A math sentence with an equal sign showing both sides are the same.	x + 25 = 60 — both sides equal 60 when x = 35
Inverse operation *Spanish: Operación inversa*	Two math actions that undo each other, like × and ÷.	+ and −
Isolate *Spanish: Despejar*	To get the letter by itself on one side.	get x by itself
Solution *Spanish: Solución*	A number that makes the equation or inequality true.	x = 7 is the solution to x + 5 = 12 because 7 + 5 = 12 ✓
One-step equation *Spanish: Ecuación de un paso*	An equation you can solve in a single step using one inverse operation.	x + 7 = 20
Balance *Spanish: Equilibrio*	Keeping both sides of an equation equal by doing the same thing to each side.	do the same to each side

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A team had some points, then scored 14 more and has 40. Solve y + 14 = 40.

1. Subtract 14 from both sides: y + 14 − 14 = 40 − 14.
2. Simplify: y = 26.
3. Check: 26 + 14 = 40.

Answer: y = 26 points

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Solve x − 4 = 9. Is x being added or subtracted?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The variable has _____ added to it, so I _____ from both sides.
- I must do the same thing to both sides because _____ .
- My solution is _____ , and my check shows _____ .
- Addition and subtraction are inverse operations because _____ .

Word bank equation | variable | inverse operation | subtract | isolate | balance | both sides | solution | check | undo | Property of Equality

WATCH OUT!

A common mistake

A common mistake is applying the same operation instead of the inverse

LESSON 8.2

Write and Solve Equations Using Addition or Subtraction

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Use the number line to solve x + 8 = 20. Start at 20 and count back 8 to find x.



HINT Start at 20 on the number line. Since x + 8 = 20, subtract 8 to find x.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Solve: x + 9 = 15

(A) x = 6
(B) x = 9
(C) x = 15
(D) x = 24

HINT x has 9 added to it. To get x alone, you need to UNDO that addition.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: $y - 7 = 20$

(A) $y = 7$

(B) $y = 13$

(C) $y = 27$

(D) $y = 140$

HINT Look at what's happening to y: 7 is being SUBTRACTED. Which operation undoes subtraction?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Solve: $n + 15 = 42$

(A) $n = 15$

(B) $n = 27$

(C) $n = 42$

(D) $n = 57$

HINT n is being increased by 15. To find n alone, you must undo that addition.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Solve: $m - 11 = 25$

(A) $m = 14$

(B) $m = 25$

(C) $m = 36$

(D) $m = 275$

HINT m has 11 taken away from it. Think about which operation puts that 11 back.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Equation** $x + 6 = 14$
2. **Choose inverse** The inverse of $+ 6$ is $- 6$.
3. **Subtract from one side only** $x + 6 - 6 = 14$, so $x = 14$

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Step 3 only subtracted 6 from the left side. What happened to the right side?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why do you have to subtract 19 from BOTH sides, not just one?

SENTENCE STARTER I subtract from both sides because an equation is like a ___ — to keep it balanced, any operation on one side must ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 8.2

Write and Solve Equations Using Addition or Subtraction

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Decide whether each pair balances. Check Yes or No, then explain.**

Solve: m − 14 = 32

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

2 **Complete the table. Show how you found each value.**

Complete the table: identify the inverse operation, solve, and check.

Equation	Inverse Operation	Solution	Check
x + 18 = 45			
p − 22 = 19			
w + 3.5 = 10			

SHOW YOUR WORK

3 Write the name of the correct group on each line.

Sort each equation by the inverse operation needed to solve it.

Groups: Add to both sides | Subtract from both sides

_____ x − 9 = 14

_____ n − 20 = 35

_____ y − 6 = 8

_____ x + 12 = 30

_____ n + 7 = 19

_____ y + 15 = 40

PART 2 Apply it

real-world problems

4 Circle the letter of the best answer. Show how you know.

A shirt costs $d. After a $15 discount, it costs $38. What is d?

(A) $23

(B) $38

(C) $45

(D) $53

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Equation** y − 13 = 28
2. **Choose inverse** The inverse of − 13 is + 13, so add 13 to both sides.
3. **Apply to both sides** y = 28 + 13
4. **Simplify** y = 31

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why do you have to subtract 19 from BOTH sides, not just one?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 8.2

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Challenge · Extension

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PART 1 Understand the idea

models and meaning

1 **Decide whether each pair balances. Check Yes or No, then explain.**

Solve using decimals: x + 2.7 = 9.3. Use inverse operations.

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

_____

_____

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find the Equation Error

1. **Equation** x − 18 = 45
2. **Identify inverse** Inverse of subtraction is subtraction
3. **Apply to both sides** x − 18 − 18 = 45 − 18
4. **Simplify** x = 27

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

_____

_____

_____

_____

Correct answer: _____

3 **Answer in complete sentences.**

A detective solved x + 25 = 60 and got x = 35. Another detective solved y − 25 = 60 and got y = 35. Are both correct? Explain using inverse operations and check each answer.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Why do you have to subtract 19 from BOTH sides, not just one?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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